

Research Backup Architecture: Ongoing System and GitHub Archival

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Version control is the starting point; a backup architecture is the surrounding structure that makes it durable.

1 Introduction

GitHub alone is not a backup. That claim may sound counterintuitive to those who have been committing and pushing faithfully for years, but it describes a real architectural gap. GitHub is a remote source-of-truth tier, not a backup tier: it holds one copy, operated by one company, under one pricing model, with no local fallback. A suspended account, a platform outage, or a credential compromise can sever access immediately.

The same gap appears at the local level. Manual `git push` commands get skipped during busy stretches. An external drive's Time Machine backup silently stops when the drive is not connected. Cloud sync replicates files but does not preserve commit history. Each tool covers one slice of the risk surface; none covers it entirely.

In this post we bring together two complementary approaches to research backup:

1. **An ongoing three-tier system** that runs automatically every 15 minutes, pushing all dirty Git repositories to GitHub, synchronising files to cloud storage, and relying on Time Machine for system-wide safety.
2. **A bulk GitHub archival procedure** that creates verified local mirrors of 400+ private repositories, exports all GitHub-side metadata (issues, PRs, releases, wikis), and supports selective deletion after confirmed verification.

Together, these two components form a complete backup architecture: the ongoing system keeps the day-to-day workflow protected, and the archival procedure handles the periodic task of mirroring GitHub itself, so that the source-of-truth tier survives the loss of the account or platform.

We note that both components document the backup layer of the Workflow Construct described in [post 52](#). Post 52 names backup as a load-bearing layer under the principle ‘two tiers or it is not backup’; the configuration here implements three active tiers for daily use, plus a fourth archival leaf that mirrors GitHub off-platform.

1.1 Motivations

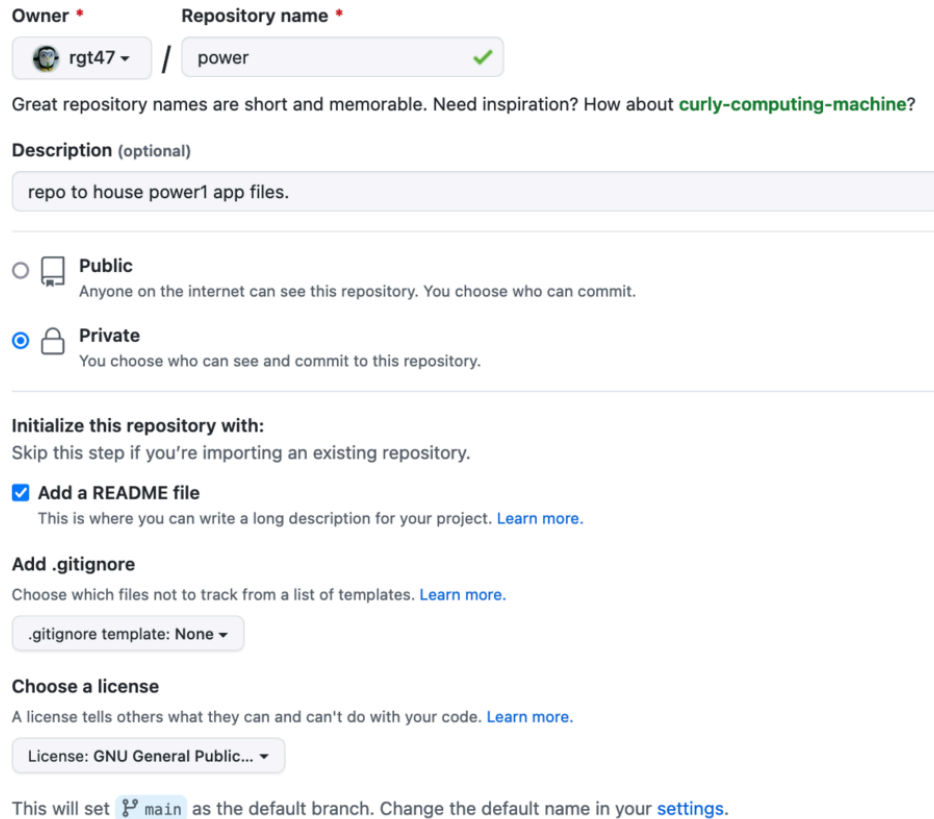
The following considerations motivated this architecture:

- Having 400+ private repositories on GitHub with no local mirrors creates a single point of failure for years of accumulated work.
- Running `git push` manually and forgetting for days at a time leaves important work vulnerable to local disk failure.
- A plain `git clone` captures code history but misses issues, pull requests, releases, and wiki content that can be more valuable than the code itself.
- A colleague’s hard-drive failure, which erased months of analytical work, demonstrated that a single backup tier is never sufficient.
- Cloud synchronisation provides file-level replication but does not preserve Git commit history or GitHub metadata.
- An automated solution must handle hundreds of repositories without manual intervention for each one.
- The archival script requires a dry-run mode so that every action can be previewed before anything destructive is attempted.

1.2 Objectives

1. Configure Time Machine as a system-wide safety net for files that live outside Git.
2. Write and schedule a production-grade script that commits and pushes all dirty Git repositories automatically every 15 minutes.
3. Build a bulk archival script that creates a full mirror, portable bundle, wiki clone, and metadata export for every private repository on a GitHub account.
4. Implement a verification phase that checks every bundle with `git bundle verify` before any deletion is permitted.
5. Add a selective preservation mechanism so active or shared repos remain on GitHub while dormant ones are archived and removed.

Corrections and alternative approaches are welcome.



The screenshot shows the GitHub repository creation interface. At the top, the 'Owner' is 'rgt47' and the 'Repository name' is 'power', which is marked as valid with a green checkmark. Below this, a suggestion for repository names is provided: 'Great repository names are short and memorable. Need inspiration? How about curly-computing-machine?'. The 'Description' field is optional and contains the text 'repo to house power1 app files.'. The repository is set to 'Private', with a note that the user chooses who can see and commit to it. Under 'Initialize this repository with:', the 'Add a README file' option is checked, with a note that this is where a long description for the project can be written. The 'Add .gitignore' section shows a dropdown menu set to '.gitignore template: None'. The 'Choose a license' section shows a dropdown menu set to 'License: GNU General Public...'. At the bottom, it states that the default branch will be set to 'main'.

Figure 1: A workspace ready for a focused archival session.

2 Prerequisites and Setup

2.1 For the Ongoing Backup System

We assume throughout a macOS environment with the following tools available:

- **macOS** 12 (Monterey) or later
- **Homebrew Bash** (/opt/homebrew/bin/bash): macOS ships with Bash 3.2, but the scripts here use Bash 4+ features
- **Git** 2.30 or later, configured with SSH keys for GitHub
- **A USB external drive** (1 TB recommended) for Time Machine
- **A cloud sync service** (Google Drive, Dropbox, or iCloud) for the third tier

The research directory structure assumed throughout is ~/prj/, containing all Git repositories. Paths should be adjusted to match one's own directory layout.

2.2 For the GitHub Archival Script

Before running the archival script, we verify that three tools are installed and authenticated:

```
gh auth status
```

```
git --version
```

```
df -h ~
```

The disk space check is worth doing early: 400 repositories can require 20-40 GB depending on release asset history.

A GitHub personal access token with `repo` and `delete_repo` scopes is also needed for those who intend to use the deletion phase. The `gh auth login` flow can configure this interactively; for backup-only runs, the standard `repo` scope is sufficient.

3 What is a Research Backup Architecture?

A research backup architecture layers multiple independent protection mechanisms so that no single point of failure can result in data loss. We shall describe the architecture here in terms of four components:

1. **Automated Git commits and pushes** (every 15 minutes): protects against uncommitted work and local corruption by pushing changes to GitHub.
2. **Cloud synchronisation** (real-time via Google Drive or Dropbox): provides continuous file-level replication across devices, useful for non-Git files and immediate access from other machines.
3. **Time Machine backups** (hourly, system-wide): captures the entire filesystem including system settings, application data, and files not covered by the other two tiers.
4. **Periodic GitHub archival** (quarterly or before any account change): creates verified local mirrors of every private repository, exporting all GitHub-side metadata that a plain `git clone` would miss: issue threads, pull request discussions, release binaries, labels, milestones, and wikis.

Each tier compensates for a weakness in the others. Git does not capture large binary files well; cloud sync does not preserve commit history; Time Machine does not push data off-site; GitHub holds only one copy of the data it hosts. Together, the four components cover the full risk surface.

4 Section 1: The Three-Tier Ongoing Backup System

4.1 Configuring Time Machine

Time Machine provides system-wide backup protection and serves as the safety net for everything beyond Git repositories.

4.1.1 Connect and Format the USB Drive

1. Connect your USB drive to the MacBook.
2. When prompted, do not use it for Time Machine yet; configure it properly first.
3. Open Disk Utility (Applications > Utilities > Disk Utility).
4. Select the USB drive from the sidebar.
5. Click Erase.
6. Choose format: APFS (recommended for modern Macs) or Mac OS Extended (Journaled).
7. Name it something recognisable, such as 'Research Backup'.
8. Click Erase.

4.1.2 Configure Time Machine

1. Open System Preferences > Time Machine.
2. Click Select Backup Disk.
3. Choose your USB drive.
4. Click Use Disk.
5. If prompted about encryption, choose Encrypt Backup for security.

4.1.3 Customise Exclusions

1. Click Options in Time Machine preferences.
2. Add folders to exclude: Downloads, Trash, virtual machines, and similar high-churn directories.
3. Do not exclude ~/prj – that directory should be covered as a secondary layer behind Git.
4. Enable 'Back up while on battery power' if you work unplugged frequently.

Time Machine will now back up the entire system (including ~/prj) every hour when the USB drive is connected.

4.2 The Minimal Backup Script

Before presenting the full production script, it is useful to see the core logic in its simplest form. This minimal version does three things: it finds every Git repository under ~/prj, checks whether that repository has uncommitted changes, and pushes those changes to the remote.

```
#!/opt/homebrew/bin/bash

find "$HOME/prj" -name ".git" -type d \
  | while read git_dir; do
    cd "$(dirname "$git_dir")" || continue
    [[ -n $(git status --porcelain) ]] || continue
    git add -A
    git commit -m \
      "Auto-backup: $(date '+%Y-%m-%d %H:%M:%S')"
```

```
git push origin main 2>/dev/null \  
|| git push origin master 2>/dev/null  
done
```

This works, but it lacks error handling, logging, user filtering, and any mechanism for diagnosing failures. The full script below addresses each of these gaps.

4.3 The Full Backup Script

The production script extends the minimal version with comprehensive features. The sections below walk through it in logical segments.

4.3.1 Configuration and Argument Parsing

```
#!/opt/homebrew/bin/bash  
  
RESEARCH_DIR="$HOME/prj/"  
LOG_FILE="$HOME/Library/Logs/research_backup.log"  
MAX_LOG_SIZE=10485760  
VERBOSE=false  
  
while [[ $# -gt 0 ]]; do  
    case $1 in  
        -v|--verbose)  
            VERBOSE=true  
            shift  
            ;;  
        -h|--help)  
            echo "Usage: $0 [-v|--verbose] \  
                [-h|--help]"  
            echo "  -v, --verbose \  
                Enable verbose output"  
            echo "  -h, --help \  
                Show this help message"  
            exit 0  
            ;;  
        *)  
            echo "Unknown option: $1"  
            echo "Use -h or --help for usage"  
            exit 1  
            ;;  
    esac  
done
```


4.3.2 Logging Infrastructure

Log rotation prevents the log file from growing without bound. The `log_message` function writes every event to disk and optionally echoes colour-coded output to the console when verbose mode is active.

```
mkdir -p "$(dirname "$LOG_FILE")"

if [[ -f "$LOG_FILE" \
    && $(stat -f%z "$LOG_FILE") \
    -gt $MAX_LOG_SIZE ]]; then
    mv "$LOG_FILE" "${LOG_FILE}.old"
    if [[ "$VERBOSE" == true ]]; then
        echo "INFO: Rotated log file" \
            "(exceeded ${MAX_LOG_SIZE} bytes)"
    fi
fi

log_message() {
    local level="$1"
    local message="$2"
    local timestamp
    timestamp=$(date '+%Y-%m-%d %H:%M:%S')
    local log_entry
    log_entry="$timestamp: [$level] $message"

    echo "$log_entry" >> "$LOG_FILE"

    if [[ "$VERBOSE" == true ]]; then
        case "$level" in
            ERROR)
                echo -e \
                    "\033[31m$log_entry\033[0m"
                ;;
            WARNING)
                echo -e \
                    "\033[33m$log_entry\033[0m"
                ;;
            SUCCESS)
                echo -e \
                    "\033[32m$log_entry\033[0m"
                ;;
            INFO)
                echo -e \
                    "\033[34m$log_entry\033[0m"
                ;;
            *)

```

```

        echo "$log_entry"
        ;;
    esac
fi
}

```

4.3.3 Repository Validation Functions

Four helper functions handle the filtering logic. `check_remote` verifies that the repository has an origin remote. `check_user_association` ensures that only repositories belonging to the 'rgt47' account are processed, preventing the script from pushing to collaborator remotes. `should_exclude_directory` skips archive and backup folders. `get_current_branch` and `branch_exists_on_remote` support intelligent push behaviour.

```

check_remote() {
    local repo_dir="$1"
    cd "$repo_dir" || return 1
    local remote_url
    remote_url=$(git remote get-url origin \
        2>/dev/null)
    [[ -n "$remote_url" ]]
}

check_user_association() {
    local repo_dir="$1"
    cd "$repo_dir" || return 1

    local remote_url
    remote_url=$(git remote get-url origin \
        2>/dev/null)
    if [[ "$remote_url" == *"rgt47"* ]]; then
        return 0
    fi

    local git_user git_email
    git_user=$(git config user.name 2>/dev/null)
    git_email=$(git config user.email 2>/dev/null)

    if [[ "$git_user" == *"rgt47"* ]] \
        || [[ "$git_email" == *"rgt47"* ]]; then
        return 0
    fi

    if [[ -z "$git_user" ]]; then
        git_user=$(git config --global \
            user.name 2>/dev/null)
    fi
}

```

```
fi
if [[ -z "$git_email" ]]; then
    git_email=$(git config --global \
        user.email 2>/dev/null)
fi

if [[ "$git_user" == *"rgt47"* ]] \
    || [[ "$git_email" == *"rgt47"* ]]; then
    return 0
fi

return 1
}

should_exclude_directory() {
    local repo_name="$1"
    local repo_path="$2"

    local lower_name lower_path
    lower_name=$(echo "$repo_name" \
        | tr '[:upper:]' '[:lower:]')
    lower_path=$(echo "$repo_path" \
        | tr '[:upper:]' '[:lower:]')

    if [[ "$lower_name" == *"archive"* ]] \
        || [[ "$lower_name" == *"backup"* ]]; then
        return 0
    fi

    if [[ "$lower_path" == *"archive"* ]] \
        || [[ "$lower_path" == *"backup"* ]]; then
        return 0
    fi

    return 1
}

get_current_branch() {
    git symbolic-ref --short HEAD 2>/dev/null \
        || git rev-parse --short HEAD 2>/dev/null
}

branch_exists_on_remote() {
    local branch="$1"
    git ls-remote --heads origin "$branch" \
        2>/dev/null | grep -q "$branch"
}
```

```
}
```

4.3.4 Main Loop: Discovery and Processing

The main loop uses `find` with null-delimited output to safely handle repository paths that contain spaces. Each repository passes through a series of checks before any Git operations are attempted.

```
log_message "INFO" \
    "Starting research backup scan" \
    " with verbose=$VERBOSE"

if [[ ! -d "$RESEARCH_DIR" ]]; then
    log_message "ERROR" \
        "Research directory $RESEARCH_DIR" \
        " does not exist"
    exit 1
fi

log_message "INFO" \
    "Scanning: $RESEARCH_DIR"

repo_count=0
backup_count=0
error_count=0
warning_count=0
skipped_count=0
excluded_count=0

while IFS= read -r -d '' git_dir; do
    repo_dir=$(dirname "$git_dir")
    repo_name=$(basename "$repo_dir")
    relative_path="${repo_dir#$RESEARCH_DIR}"

    if should_exclude_directory \
        "$repo_name" "$relative_path"; then
        log_message "INFO" \
            "Excluding (archive/backup):" \
            " $relative_path"
        ((excluded_count++))
        continue
    fi

    log_message "INFO" \
        "Processing: $relative_path"

    if ! cd "$repo_dir"; then
```

```
        log_message "ERROR" \  
            "Cannot access: $repo_dir" \  
        ((error_count++)) \  
        continue \  
    fi \  
 \  
    ((repo_count++)) \  
 \  
    if ! git rev-parse --git-dir \  
        >/dev/null 2>&1; then \  
        log_message "ERROR" \  
            "Not a valid git repo:" \  
            " $relative_path" \  
        ((error_count++)) \  
        continue \  
    fi \  
 \  
    if ! check_user_association "$repo_dir"; then \  
        log_message "INFO" \  
            "Skipping (not rgt47):" \  
            " $relative_path" \  
        ((skipped_count++)) \  
        continue \  
    fi \  
 \  
    log_message "INFO" \  
        "$relative_path associated with rgt47" \  
 \  
    if ! check_remote "$repo_dir"; then \  
        log_message "WARNING" \  
            "No remote configured:" \  
            " $relative_path" \  
        ((warning_count++)) \  
        ((skipped_count++)) \  
        continue \  
    fi \  
 \  
    current_branch=$(get_current_branch) \  
    if [[ -z "$current_branch" ]]; then \  
        log_message "ERROR" \  
            "Cannot determine branch:" \  
            " $relative_path" \  
        ((error_count++)) \  
        continue \  
    fi
```

```
log_message "INFO" \  
    "$relative_path on branch:" \  
    " $current_branch"  
  
git_status=$(git status --porcelain \  
    2>/dev/null)  
  
if [[ -z "$git_status" ]]; then  
    log_message "INFO" \  
        "$relative_path is clean"  
    continue  
fi  
  
untracked=$(echo "$git_status" \  
    | grep -c "^?" || echo 0)  
modified=$(echo "$git_status" \  
    | grep -c "^ M" || echo 0)  
added=$(echo "$git_status" \  
    | grep -c "^A " || echo 0)  
deleted=$(echo "$git_status" \  
    | grep -c "^D " || echo 0)  
  
log_message "INFO" \  
    "$relative_path: $untracked new," \  
    " $modified modified, $added added," \  
    " $deleted deleted"  
  
if ! git add -A 2>/dev/null; then  
    log_message "ERROR" \  
        "Failed to stage: $relative_path"  
    ((error_count++))  
    continue  
fi  
  
log_message "INFO" \  
    "Staged changes: $relative_path"  
  
commit_message="Auto-backup:" \  
    " $(date '+%Y-%m-%d %H:%M:%S')"  
  
if git commit -m "$commit_message" \  
    >/dev/null 2>&1; then  
    log_message "SUCCESS" \  
        "Committed: $relative_path"  
  
    if ! branch_exists_on_remote \  

```

```
    "$current_branch"; then
log_message "WARNING" \
    "'$current_branch' not on" \
    " remote: $relative_path"

if git push --set-upstream origin \
    "$current_branch" 2>/dev/null
then
    log_message "SUCCESS" \
        "Created and pushed" \
        " '$current_branch':" \
        " $relative_path"
    ((backup_count++))
else
    log_message "ERROR" \
        "Failed to push new" \
        " branch: $relative_path"
    ((error_count++))
fi
else
if git push origin \
    "$current_branch" 2>/dev/null
then
    log_message "SUCCESS" \
        "Pushed '$current_branch':" \
        " $relative_path"
    ((backup_count++))
else
    log_message "ERROR" \
        "Push failed:" \
        " $relative_path" \
        " (check network/auth)"
    ((error_count++))
fi
fi
else
if git diff --cached --quiet; then
    log_message "INFO" \
        "No changes to commit:" \
        " $relative_path"
else
    log_message "ERROR" \
        "Commit failed:" \
        " $relative_path"
    ((error_count++))
fi
fi
```

```
done < <(find "$RESEARCH_DIR" \  
-name ".git" -type d -print0)
```

4.3.5 Summary Report

After processing every repository, the script logs aggregate statistics and, in verbose mode, prints a human-readable summary to the console.

```
log_message "INFO" "Backup scan complete"  
log_message "INFO" \  
    "Summary: $repo_count processed," \  
    " $backup_count backed up"  
log_message "INFO" \  
    "Excluded: $excluded_count," \  
    " Skipped: $skipped_count," \  
    " Errors: $error_count," \  
    " Warnings: $warning_count"  
  
if [[ "$VERBOSE" == true ]]; then  
    echo ""  
    echo "=== BACKUP SUMMARY ==="  
    echo "Repositories found:" \  
        "$((repo_count + excluded_count" \  
        " + skipped_count))"  
    echo "Excluded: $excluded_count" \  
        "(archive/backup)"  
    echo "Skipped: $skipped_count (not rgt47)"  
    echo "Processed: $repo_count"  
    echo "Backed up: $backup_count"  
    echo "Errors: $error_count"  
    echo "Warnings: $warning_count"  
    echo ""  
    echo "Log file: $LOG_FILE"  
  
    if [[ $error_count -gt 0 ]]; then  
        echo ""  
        echo "WARNING: There were errors" \  
            "during backup. Check the log" \  
            "file for details."  
        exit 1  
    elif [[ $warning_count -gt 0 ]]; then  
        echo ""  
        echo "NOTE: Backup completed with" \  
            "warnings. Check the log file" \  
            "for details."
```



```
    else
        echo ""
        echo "Backup completed successfully."
    fi
fi

exit 0
```

4.4 Scheduling with Cron

The final step is to make the script run automatically. A cron job at 15-minute intervals provides a good balance between backup frequency and system resource usage.

```
crontab -e
```

Add the following entry:

```
*/15 * * * * /Users/${whoami}/scripts/backup-research.sh
```

Save and exit (Ctrl+X, then Y, then Enter in nano; or Esc, :wq, Enter in vim). Verify:

```
crontab -l
```

Wait 15 minutes, then confirm execution by inspecting the log:

```
tail -20 ~/Library/Logs/research_backup.log
```



Figure 2: Terminal windows and backup logs – the machinery behind automated research protection.

5 Section 2: Bulk GitHub Account Archival

The ongoing backup system protects daily work, but it does not address the accumulation of private repositories that exist only on GitHub. A GitHub archive, in this context, is a local copy of everything GitHub stores for a repository: not just the code and commit history, but also the metadata that lives only on GitHub’s servers (issue threads, pull request discussions, release binaries, labels, milestones, and wiki pages).

A regular `git clone` captures the commit graph but misses all of that surrounding context. A mirror clone (`git clone --mirror`) captures every ref, including remote-tracking branches and tags. A git bundle packages that mirror into a single portable file. And the GitHub API exports capture the metadata that git itself does not track.

The archival script combines all three approaches: mirror clone for completeness, bundle for portability, and API exports for metadata. The result is a self-contained backup directory per repository that can survive even if GitHub itself becomes unavailable.

5.1 The Three-Phase Approach

The archive script follows a strict three-phase process. Understanding this structure makes the full script easier to follow.

Phase 1: Backup Everything. For each repository, the script creates:

- A full git mirror with every branch and tag
- A portable bundle file for easy transfer
- Wiki content (if the repo has one)
- Metadata exports via the GitHub API (issues, PRs, releases, labels, milestones, workflows)
- Downloaded release assets (binaries, artifacts)

Phase 2: Verify Backups. Before any deletion, the script runs `git bundle verify` on every bundle. If any verification fails, the entire deletion phase is aborted. This step is essential: without it, the deletion phase cannot be trusted.

Phase 3: Selective Deletion. Only repos not in the keep-list get deleted, and only after an explicit typed confirmation (DELETE). Repos in `KEEP_ON_GITHUB` are still backed up but are not removed from GitHub.

5.2 The Complete Working Script

Save the full script as `github-archive.sh` and make it executable with `chmod +x github-archive.sh`.

5.2.1 Script Header and Configuration

```
#!/bin/bash

set -e

OWNER="your-username"
BACKUP_DIR="$HOME/github-archive"
DATE=$(date +%Y%m%d)
LOG_FILE="$BACKUP_DIR/archive_$(date +%Y%m%d).log"
DRY_RUN=false

KEEP_ON_GITHUB=(
    "important-project"
    "active-work"
    "shared-with-team"
)

usage() {
    echo "Usage: $0 [OPTIONS]"
    echo ""
```

```

    echo "Options:"
    echo "  -n, --dry-run      Show what would happen"
    echo "  -o, --owner NAME   GitHub username/org"
    echo "  -d, --dir PATH     Backup directory"
    echo "  -h, --help         Show this help message"
    exit 0
}

while [[ $# -gt 0 ]]; do
    case $1 in
        -n|--dry-run)
            DRY_RUN=true
            shift
            ;;
        -o|--owner)
            OWNER="$2"
            shift 2
            ;;
        -d|--dir)
            BACKUP_DIR="$2"
            shift 2
            ;;
        -h|--help)
            usage
            ;;
        *)
            echo "Unknown option: $1"
            usage
            ;;
    esac
done

mkdir -p "$BACKUP_DIR"

```

The `KEEP_ON_GITHUB` array is the most important configuration. Reviewing the repository list before running the script ensures the right ones are included.

5.2.2 Utility Functions

```

log() {
    local prefix=""
    if [ "$DRY_RUN" = true ]; then
        prefix="[DRY-RUN] "
    fi
    echo "[$(date '+%Y-%m-%d %H:%M:%S')] \
${prefix}$1" | tee -a "$LOG_FILE"
}

```

```
}

is_kept() {
    local repo=$1
    for kept in "${KEEP_ON_GITHUB[@]}; do
        if [ "$repo" = "$kept" ]; then
            return 0
        fi
    done
    return 1
}
```

5.2.3 Phase 1: The Backup Function

```
backup_repo() {
    local repo=$1
    local repo_dir="$BACKUP_DIR/$repo"

    log "=== Backing up $repo ==="

    if [ "$DRY_RUN" = true ]; then
        log "Would create: $repo_dir"
        log "Would clone: $OWNER/$repo"
        log "Would export: issues, PRs, releases"
        return
    fi

    mkdir -p "$repo_dir"
    cd "$repo_dir"

    if [ ! -d "repo.git" ]; then
        log "Cloning repository..."
        gh repo clone "$OWNER/$repo" \
            repo.git -- --mirror
    else
        log "Updating existing clone..."
        cd repo.git && git fetch --all && cd ..
    fi

    log "Creating git bundle..."
    cd repo.git
    git bundle create ../repo.bundle --all
    cd ..

    log "Checking for wiki..."
```

```

if gh api "repos/$OWNER/$repo" \
  --jq '.has_wiki' 2>/dev/null \
  | grep -q true; then
  git clone \
    "https://github.com/$OWNER/$repo.wiki.git" \
    wiki.git 2>/dev/null \
    || log "No wiki content"
fi

log "Exporting metadata..."
gh api "repos/$OWNER/$repo" \
  > repo-info.json 2>/dev/null || true
gh api "repos/$OWNER/$repo/issues?state=all" \
  --paginate > issues.json 2>/dev/null || true
gh api "repos/$OWNER/$repo/pulls?state=all" \
  --paginate \
  > pull-requests.json 2>/dev/null || true
gh api "repos/$OWNER/$repo/releases" \
  --paginate > releases.json 2>/dev/null || true
gh api "repos/$OWNER/$repo/labels" \
  --paginate > labels.json 2>/dev/null || true
gh api "repos/$OWNER/$repo/milestones?state=all" \
  --paginate \
  > milestones.json 2>/dev/null || true

if [ -s releases.json ] \
  && [ "$(cat releases.json)" != "[]" ]; then
  log "Downloading release assets..."
  mkdir -p release-assets
  gh release list -R "$OWNER/$repo" \
    --limit 100 2>/dev/null \
    | while read -r tag rest; do
    gh release download "$tag" \
      -R "$OWNER/$repo" \
      -D "release-assets/$tag" \
      2>/dev/null || true
  done
fi

log "Completed backup of $repo"
cd "$BACKUP_DIR"
}

```

We note that the `--paginate` flag on `gh api` calls is essential. Without it, GitHub's API returns only the first 30 items per endpoint, with the consequence that issues and pull requests on larger repositories are silently lost.

5.2.4 Phase 2: The Verification Function

```
verify_backup() {
    local repo=$1
    local repo_dir="$BACKUP_DIR/$repo"

    log "Verifying backup of $repo..."

    if [ "$DRY_RUN" = true ]; then
        log "Would verify: $repo_dir/repo.bundle"
        return 0
    fi

    if [ -f "$repo_dir/repo.bundle" ]; then
        cd "$repo_dir"
        if git bundle verify repo.bundle \
            > /dev/null 2>&1; then
            log "PASS: Bundle verified"
            return 0
        else
            log "FAIL: Bundle verification failed!"
            return 1
        fi
    else
        log "FAIL: Bundle not found!"
        return 1
    fi
}
```

5.2.5 Phase 3: The Deletion Function

```
delete_repo() {
    local repo=$1

    if [ "$DRY_RUN" = true ]; then
        log "Would delete: $OWNER/$repo"
        return
    fi

    log "Deleting $repo from GitHub..."
    gh repo delete "$OWNER/$repo" --yes
    log "Deleted $repo"
}
```

5.2.6 Main Execution Flow

```

if [ "$DRY_RUN" = true ]; then
    echo "=====
    echo " DRY-RUN MODE - No changes will be made"
    echo "=====
    echo ""
fi

log "Starting GitHub archive process"
log "Owner: $OWNER"
log "Backup directory: $BACKUP_DIR"

log "Fetching list of private repositories..."
repos=$(gh repo list "$OWNER" \
    --limit 500 --private \
    --json name -q '.[].name')
repo_count=$(echo "$repos" | wc -l | tr -d ' ')
log "Found $repo_count private repositories"

repos_to_delete=""
repos_to_keep=""
delete_count=0
keep_count=0

for repo in $repos; do
    if is_kept "$repo"; then
        repos_to_keep="$repos_to_keep $repo"
        ((keep_count++)) || true
    else
        repos_to_delete="$repos_to_delete $repo"
        ((delete_count++)) || true
    fi
done

log "Repos to archive and DELETE: $delete_count"
log "Repos to archive and KEEP: $keep_count"

echo ""
echo "=== REPO CATEGORIZATION ==="
echo ""
echo "Will be DELETED after backup ($delete_count):"
for repo in $repos_to_delete; do
    echo "  x $repo"
done
echo ""

```



```

echo "Will be KEPT on GitHub ($keep_count):"
for repo in $repos_to_keep; do
    echo "  + $repo"
done
echo ""

if [ "$DRY_RUN" = true ]; then
    echo "=== DRY-RUN: PHASE 1 (BACKUP) ==="
    for repo in $repos; do
        backup_repo "$repo"
    done

    echo ""
    echo "=== DRY-RUN: PHASE 2 (VERIFICATION) ==="
    for repo in $repos; do
        verify_backup "$repo"
    done

    echo ""
    echo "=== DRY-RUN: PHASE 3 (DELETION) ==="
    for repo in $repos_to_delete; do
        delete_repo "$repo"
    done

    echo ""
    echo "===== "
    echo "  DRY-RUN COMPLETE"
    echo "===== "
    echo ""
    echo "Run without --dry-run to execute."
    exit 0
fi

log "=== PHASE 1: BACKUP ==="
for repo in $repos; do
    backup_repo "$repo"
done

log "=== PHASE 2: VERIFICATION ==="
failed_repos=""
for repo in $repos; do
    if ! verify_backup "$repo"; then
        failed_repos="$failed_repos $repo"
    fi
done

```

```
if [ -n "$failed_repos" ]; then
    log "WARNING: Verification failed:$failed_repos"
    log "Aborting deletion phase"
    exit 1
fi

log "All backups verified successfully!"

log "=== PHASE 3: DELETION ==="
echo ""
echo "Backup complete and verified!"
echo ""
read -p "Delete $delete_count repos? \
(type 'DELETE'): " confirm

if [ "$confirm" = "DELETE" ]; then
    for repo in $repos_to_delete; do
        delete_repo "$repo"
    done
    log "Deleted $delete_count repositories"
else
    log "Deletion cancelled"
fi

log "Archive process complete"
```

5.3 Using the Archival Script

```
./github-archive.sh --dry-run

./github-archive.sh

./github-archive.sh \
    --owner myorg --dir /external/drive/backup
```

Running the dry-run first is not merely useful; it is essential, since the categorisation output can catch repositories that were inadvertently omitted from the keep-list.

Configure the `KEEP_ON_GITHUB` array to match the deployment:

```
KEEP_ON_GITHUB=(
    "active-projects"
    "shared-with-team"
    "client-work"
    "portfolio"
)
```

5.4 Backup Directory Structure

After running the script, the backup directory looks like this:

```
~/github-archive/
+-- important-project/      # KEPT on GitHub
|   +-- repo.git/
|   +-- repo.bundle
|   +-- repo-info.json
|
+-- old-project-1/         # DELETED from GitHub
|   +-- repo.git/
|   +-- repo.bundle
|   +-- wiki.git/
|   +-- repo-info.json
|   +-- issues.json
|   +-- pull-requests.json
|   +-- releases.json
|   +-- labels.json
|   +-- milestones.json
|   +-- release-assets/
|
+-- archive_20260517.log
```

Content	Format	Use Case
Git history	repo.git/ + bundle	Full reproducibility
Wiki	wiki.git/	Documentation
Issues	issues.json	Discussion archive
Pull requests	pull-requests.json	Code review history
Releases	releases.json	Version history
Release assets	release-assets/	Binaries, artifacts
Metadata	repo-info.json	Repository config
Labels	labels.json	Issue classification
Milestones	milestones.json	Project tracking

5.5 Restoring from Archive

```
git clone \
  ~/github-archive/repo-name/repo.bundle \
  restored-repo

gh repo create new-repo-name --private
cd ~/github-archive/repo-name/repo.git
git push --mirror \
```

```
git@github.com:you/new-repo-name.git
```

The bundle approach is more convenient for quick local inspection; the mirror push is better for actually recreating a repository on GitHub.

6 Section 3: Verification and Testing

6.1 Ongoing Backup Verification

To confirm that the daily backup script is running correctly:

```
tail -20 ~/Library/Logs/research_backup.log
```

```
~/scripts/backup-research.sh --verbose
```

Check that recent log entries show [SUCCESS] for pushed repositories and that the timestamp reflects the expected cron interval.

6.2 GitHub Archive Verification

Before trusting any archival backup, run these checks:

```
cd ~/github-archive/repo-name
```

```
git bundle verify repo.bundle
```

```
ls -la *.json
```

```
git bundle list-heads repo.bundle
```

```
python3 -m json.tool issues.json | head -50
```

These checks confirm that the bundle is structurally valid, that metadata files are non-empty, that all branches are present, and that the issue export is well-formed JSON.



Figure 3: Organised workspace: the goal after archiving 400 repos.

After the archive, what remains on GitHub is clean and intentional.

7 Daily Workflow

Command	Action
<code>~/scripts/backup-research.sh --verbose</code>	Run ongoing backup with console output
<code>tail -20 ~/Library/Logs/research_backup.log</code>	Review recent backup log entries
<code>bash github-archive.sh --dry-run</code>	Preview what would be backed up or deleted
<code>bash github-archive.sh</code>	Run full archival: backup, verify, confirm deletion
<code>bash github-archive.sh --owner ORG</code>	Archive a different owner's repositories
<code>git bundle verify repo.bundle</code>	Spot-check a specific archival bundle
<code>gh repo list --json name</code>	Confirm which repositories remain on GitHub
<code>crontab -l</code>	Verify that the 15-minute cron job is registered

Run the archival script quarterly or before any GitHub plan or account change. The ongoing backup script runs automatically once scheduled.

8 Things to Watch Out For

1. **Homebrew Bash path.** macOS ships with Bash 3.2, which lacks features the backup script depends on. Ensure the shebang points to `/opt/homebrew/bin/bash` (Apple Silicon) or `/usr/local/bin/bash` (Intel). Version mismatches can cause silent failures that are difficult to diagnose.
2. **SSH key agent in cron.** Cron jobs do not inherit your shell environment. If Git remotes use SSH, the cron job may fail silently because `ssh-agent` is not available. Add `eval "$(ssh-agent -s)"` to the script or use macOS Keychain integration.
3. **The `--limit 500` cap on `gh repo list`.** The command defaults to 30 results. The archival script sets it to 500, but if you have more repositories than that, you need to increase the limit or paginate manually.
4. **Disk space surprises.** An initial estimate of 20 GB for 400 repos may prove insufficient; closer to 35 GB may be needed if several repos have large binary assets in their release history. Check with `df -h` before starting.
5. **Network interruptions during archival.** If a clone fails midway, the `repo.git` directory exists but is incomplete. The verification phase catches this, but you must delete the partial clone and rerun.
6. **API rate limits.** GitHub's API allows 5,000 requests per hour for authenticated users. With 400 repos and 6 API calls each, that is 2,400 requests – within the limit but close. If the limit is reached, the script pauses without automatic retry.
7. **Time Machine drive connection.** Time Machine requires the USB drive to be physically connected. When travelling without the drive, this tier is inactive. The Git push tier continues to operate but cloud sync becomes the only off-site copy.

9 Uninstall / Rollback

9.1 Removing the Ongoing Backup System

```
crontab -e
```

Delete the line referencing `backup-research.sh`. Then remove the script:

```
rm -i ~/scripts/backup-research.sh
```

The log file can be removed independently:

```
rm -i ~/Library/Logs/research_backup.log
```

9.2 Restoring from the GitHub Archive

```
cd ~/github-archive/repo-name
git clone repo.bundle restored-repo

gh repo create owner/repo-name --private
cd restored-repo
git remote add origin git@github.com:owner/repo-name.git
git push --mirror origin
```

9.3 Removing the Archive Script and Local Copies

```
rm -i github-archive.sh
rm -ri ~/github-archive/
```

Confirm before removing backups: if GitHub has already been cleaned up, the local archive is the only copy.

10 What Did We Learn?

10.1 Lessons Learnt

Conceptual Understanding:

- GitHub is a source-of-truth tier, not a backup tier. A suspended account, a pricing change, or a credential compromise can sever access immediately, regardless of how many commits are in the history.
- A single backup mechanism covers only one slice of the risk surface. The combination of automated Git pushes, cloud sync, Time Machine, and periodic archival covers the full surface more reliably than any one mechanism alone.
- A plain `git clone` misses metadata that can be more valuable than the code itself: issue discussions, PR review threads, and release notes.
- The three-phase archival pattern (backup, verify, delete) is a general discipline applicable to any destructive batch operation, not just GitHub archiving.

Technical Skills:

- Bash `find` with `-print0` and `read -d ''` safely handles directory names containing spaces and special characters, which is common in research project naming.
- The `gh` CLI is more powerful than often expected. Combining `gh repo list`, `gh api --paginate`, and `gh release download` covers nearly every GitHub operation without touching the web interface.
- `git bundle verify` is a built-in safety net that may be unfamiliar. It confirms the bundle is a valid, complete repository.

- Log rotation using file size checks (`stat -f%z`) prevents unbounded log growth in long-running automated scripts.

Gotchas and Pitfalls:

- Cron does not source `.bashrc` or `.zshrc`, so environment variables, SSH keys, and PATH modifications are not available unless explicitly set within the script or the crontab.
- The `--paginate` flag on `gh api` is critical. Forgetting it silently loses issues beyond the first 30 on larger repos.
- GitHub wikis are technically separate git repositories. They must be cloned independently, and they silently fail if the wiki was enabled but never populated.
- `set -e` causes the archival script to exit on the first error, which is good for safety but requires `|| true` on commands that are expected to fail (such as cloning an empty wiki).

10.2 Limitations

- The ongoing backup script only pushes to the `origin` remote. Repositories using multiple remotes receive backup to only one of them.
- Auto-generated commit messages ('Auto-backup: timestamp') lack descriptive content. This serves the backup purpose but pollutes Git history for active development branches.
- The archival script does not handle GitHub Actions workflow run history or Codespaces configurations, which are not available through the standard API.
- Repository settings (branch protection rules, webhook configurations, deploy keys) are not exported. Recreating those requires manual setup or additional API calls.
- Git LFS objects are not included in the mirror clone by default. Repositories using LFS need `git lfs fetch --all` added to the backup function.
- The archival script processes repositories sequentially. For 400+ repos, this can take several hours. Parallelisation would speed things up but adds complexity.
- Time Machine requires the USB drive to be physically connected. When travelling without the drive, this tier is inactive.

10.3 Opportunities for Improvement

1. Replace fixed commit messages in the ongoing backup script with a brief summary of changed file names, providing more informative Git history.
2. Add Git LFS support to the archival script by inserting `git lfs fetch --all` into the backup function for repositories that use large file storage.
3. Parallelise the archival backup phase using `xargs -P 4` or GNU `parallel` to clone multiple repositories concurrently, reducing total runtime.
4. Export repository settings (branch protection rules, webhooks, deploy keys) so that restoration is truly complete.
5. Migrate from cron to `launchd` for better macOS integration, including wake-from-sleep triggers and retry logic.
6. Add a companion `github-restore.sh` script that reads a backup directory and recreates repositories on GitHub, including metadata re-import.
7. Add compression: after archival verification, compress each repository directory with `tar -czf` to reduce storage requirements by roughly 50-70%.

11 Wrapping Up

The two components in this post address different timescales of the same underlying concern: ensuring that research work is recoverable regardless of what fails. The ongoing three-tier system handles the daily rhythm, committing and pushing every 15 minutes while Time Machine and cloud sync provide complementary coverage. The archival procedure handles the periodic task of verifying that GitHub itself (the source-of-truth tier) has a local mirror that would survive the loss of the account.

The most important lesson is not about any specific tool but about the architecture: no single backup mechanism is sufficient, verification must be part of the process rather than assumed, and automation is the only reliable way to maintain discipline across hundreds of repositories over months and years.

In conclusion, four points merit emphasis. First, GitHub is not a backup: it is one copy, held by one company, and the backup architecture is everything surrounding it. Second, three independent ongoing tiers (Git push, cloud sync, and Time Machine) cover the full spectrum of daily failure modes, and no single tier is sufficient on its own. Third, the three-phase archival pattern (backup, verify, delete) prevents data loss from incomplete archives and ensures that deletions are irreversible only after explicit confirmation. Fourth, a 400-repository archive requires roughly two to four hours and twenty to forty gigabytes of disk space, depending on the release asset history of the repositories involved.

12 See Also

Related posts on this blog:

- [Configure the Command Line for Data Science Development](#) – the terminal and shell setup that complements this backup architecture.
- [Setting Up Git for Data Science Development](#)
- [Creating a GitHub Dotfiles Repository](#)

Key resources:

- [GitHub CLI Documentation](#)
- [Git Bundle Documentation](#)
- [GitHub REST API: Repositories](#)
- [Pro Git Book \(free\)](#) – authoritative reference for Git concepts
- [Apple Time Machine Documentation](#) – official setup and troubleshooting guide
- [Crontab Guru](#) – interactive cron schedule expression editor
- [Homebrew](#) – macOS package manager for installing Bash 5 and other tools
- [Git LFS Documentation](#)

13 Reproducibility

Environment requirements:

- macOS 12 (Monterey) or later; tested on macOS 14 Sonoma
- Bash 5.x via Homebrew (macOS ships 3.2; install with `brew install bash`)
- Git 2.39 or later
- gh CLI 2.40 or later

Version checks:

```
git --version
gh --version
bash --version
```

Script files:

File	Description
backup-research.sh	Ongoing 15-minute backup
github-archive.sh	Bulk archival script
archive_YYYYMMDD.log	Archival execution log
research_backup.log	Ongoing backup log

This post does not use R or Docker. The entire workflow is pure Bash with the GitHub CLI and macOS system tools.

14 Let's Connect

Have questions, suggestions, or spot an error? Let me know.

- **GitHub:** [rgt47](#)
- **Twitter/X:** [@rgt47](#)
- **LinkedIn:** [Ronald Glenn Thomas](#)
- **Email:** [Contact form](#)

I would enjoy hearing from you if:

- You spot an error or a better approach to any of the code in this post.
- You have suggestions for topics you would like to see covered.
- You want to discuss R programming, data science, or reproducible research.
- You have questions about anything in this tutorial.
- You just want to say hello and connect.

14.1 Related posts in this cluster

This post is part of the *Security, Backup, and Sync* series. Recommended reading order:

1. **Post 31: Research Backup Architecture** (this post)
2. Post 32: [Migrating Off Dropbox: Beyond Dotfiles](#)
3. Post 33: [Setting Up pass: a Unix Password Manager](#)
4. Post 34: [Secrets Management for the Workflow Construct](#)
5. Post 35: [Security Foundations for a Multi-Laptop Research Cluster](#)